

1 / 20 | Ages 7-8 | 15 minutes

# The twenty-token shop

## What we are practising

Add prices and subtract spending from a fixed total.

## What you need

20 large paper tokens / Three household objects / Paper price labels

Draw tokens and cross them out instead of moving pieces.

## Before you begin

Label a book 7 tokens, a cup 4 and a cloth 3; allow repeated purchases.

Use large paper tokens, not real coins, when younger children are nearby.

## Try it together

1. Count a budget of 20 tokens.
2. Buy one of each item and work out the total.
3. Find the change, then choose something affordable with it.
4. Find a different basket costing exactly 20.

## Talk, draw or show your idea

Can two different baskets have the same total?

## Space to explore

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2 / 20 | Ages 7-8 | 15 minutes

## Fraction picnic

### What we are practising

Recognize equivalent shares using halves and quarters.

### What you need

Two equal rectangular sheets / Pencil

Draw two same-size rectangles and mark equal sections.

### Before you begin

Call each sheet one whole picnic sandwich.

Use paper models; no food or cutting is required.

### Try it together

1. Fold the first sheet into two equal parts and shade one.
2. Fold the second into four equal parts and shade two.
3. Overlay or line up the shaded amounts and compare.
4. Explain how two people could share one whole fairly.

### Talk, draw or show your idea

Why must the starting wholes be the same size?

### Space to explore

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3 / 20 | Ages 7-8 | 15 minutes

## Measure twice

### What we are practising

Measure a length with repeated equal units and explain unit size.

### What you need

A short pencil / A longer pencil / Paper

Use two different lengths of folded paper.

### Before you begin

Choose a reachable table edge and label the two measuring units A and B.

Measure reachable surfaces without climbing; keep pencil points away from faces.

### Try it together

1. Predict which pencil will need more end-to-end placements.
2. Move the short pencil along the edge without gaps, counting whole placements and noting any remainder.
3. Repeat with the longer pencil along the same edge.
4. Record each result with its unit name.

### Talk, draw or show your idea

Did the table change, or did the unit change?

### Space to explore

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4 / 20 | Ages 7-8 | 10 minutes

# Timetable detective

## What we are practising

Add minutes across an hour boundary.

## What you need

Paper / Pencil

Use a clock face drawn on paper.

## Before you begin

Write a start time of 3:40, followed by 15 minutes reading and 10 minutes drawing.

Keep this untimed; use invented times if schedules feel stressful.

## Try it together

1. Mark the starting time on a simple timeline.
2. Add the reading time and record its end.
3. Add the drawing time and record the final time.
4. Check by counting the total elapsed minutes.

## Talk, draw or show your idea

How many drawing minutes happen before 4:00?

## Space to explore

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5 / 20 | Ages 7-8 | 20 minutes

## A tiny survey

### What we are practising

Use tally marks and compare counts on a common scale.

### What you need

Paper / Pencil

Use an invented set of six votes instead of asking people.

### Before you begin

Choose three activity options: drawing, stories and building.

Participation is optional; record votes without names or personal details.

### Try it together

1. Ask willing family members to choose one option each, or use votes: drawing, stories, drawing, building, stories, drawing.
2. Record one tally for each vote.
3. Draw three bars with one equal-sized square per vote.
4. Label the choices and counts, then compare the tallest and shortest bars.

### Talk, draw or show your idea

Does this small survey tell us what every child prefers?

### Space to explore

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6 / 20 | Ages 7-8 | 10 minutes

## The value of the digit four

### What we are practising

Distinguish a digit from its place value.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

In the number 46, does the digit 4 mean four ones or four tens? What value does it represent?

### Space to explore

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7 / 20 | Ages 7-8 | 10 minutes

# Cross the next ten

## What we are practising

Add by bridging to a multiple of ten.

## What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

## Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

## Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

## Talk, draw or show your idea

A shelf holds 27 books. Eight more arrive. How many books are on the shelf now?

## Space to explore

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8 / 20 | Ages 7-8 | 10 minutes

## Return the borrowed books

### What we are practising

Subtract two-digit numbers across a ten.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A class borrowed 42 books and returned 17. How many are still borrowed?

### Space to explore

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9 / 20 | Ages 7-8 | 10 minutes

## Rows in the mini theatre

### What we are practising

Use multiplication for a rectangular array.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A mini theatre has four rows with six seats in every row. How many seats are there?

### Space to explore

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10 / 20 | Ages 7-8 | 10 minutes

## Three per envelope

### What we are practising

Interpret a division remainder in a packing problem.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

There are 17 cards. Each full envelope holds three cards. How many full envelopes can you make, and how many cards remain?

### Space to explore

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11 / 20 | Ages 7-8 | 10 minutes

## Which quarter amount is larger?

### What we are practising

Compare fractions with a common denominator and equal wholes.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

Two same-sized paper rectangles each have four equal parts. One has three parts coloured; the other has one. Which coloured fraction is larger?

### Space to explore

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12 / 20 | Ages 7-8 | 10 minutes

# Change from a twenty token

## What we are practising

Calculate change after combining two costs.

## What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

## Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

## Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

## Talk, draw or show your idea

At a pretend stall, a notebook costs seven tokens and a pen five. You pay twenty tokens. What change is due?

## Space to explore

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13 / 20 | Ages 7-8 | 10 minutes

## Metres and centimetres together

### What we are practising

Convert a mixed length into centimetres.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

One metre equals 100 centimetres. A ribbon is one metre and 25 centimetres long. How many centimetres is that?

### Space to explore

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14 / 20 | Ages 7-8 | 10 minutes

## A border around the card

### What we are practising

Add all side lengths to find a rectangular perimeter.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A rectangular card is eight centimetres long and three centimetres wide. How long a border goes once around all four sides?

### Space to explore

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15 / 20 | Ages 7-8 | 10 minutes

## Balance the fruit baskets

### What we are practising

Use balance equality to find an unknown mass.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

On a balanced scale, one pan holds a 500-gram weight. The other pan holds an apple weighing 200 grams and a pear. What is the pear's mass?

### Space to explore

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16 / 20 | Ages 7-8 | 10 minutes

## The workshop finish time

### What we are practising

Calculate elapsed minutes across an hour.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A workshop begins at 2:40 and lasts 35 minutes, with no break. When does it finish?

### Space to explore

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17 / 20 | Ages 7-8 | 10 minutes

# Totals from two reading days

## What we are practising

Sum selected entries from a small table described in words.

## What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

## Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

## Try it together

- 1. Read the question below and identify what you need to find.
- 2. Try an answer; show your thinking with words, fingers or a quick drawing.
- 3. Use the hint if needed, then compare with the answer and explain why it fits.

## Talk, draw or show your idea

A reading log shows Monday: 8 pages, Tuesday: 6, Wednesday: 9. How many pages were read on Monday and Wednesday together?

## Space to explore

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18 / 20 | Ages 7-8 | 10 minutes

## Even plus even

### What we are practising

Explain why a sum of two even quantities is even.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A team has six players and another has eight. If they join, can everyone form pairs with no one left over?

### Space to explore

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19 / 20 | Ages 7-8 | 10 minutes

## Halfway between two numbers

### What we are practising

Find a midpoint using equal numerical distances.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

Which whole number is halfway between 10 and 18 on a number line?

### Space to explore

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20 / 20 | Ages 7-8 | 10 minutes

## Round to the nearest ten

### What we are practising

Compare distances to adjacent multiples of ten.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A jar label says 67 beads. Rounded to the nearest ten, is that 60 or 70?

### Space to explore

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## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 1. The twenty-token shop

#### A little hint

One of each costs  $7 + 4 + 3$ ; two books leave 6 tokens.

#### Answer / what to notice

One of each costs 14, leaving 6. Two books and two cloths cost 20.

Add all prices before subtracting; a purchase is affordable if its price does not exceed the balance.

#### Make it easier

Use a budget of 10 and prices 2 and 5.

#### Take it further

Find every basket of these goods that totals 20.

#### Another day, away from the screen

Make a pretend shop with new whole-number prices.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 2. Fraction picnic

#### A little hint

Two quarters cover the same area as one half of the same whole.

#### Answer / what to notice

One half equals two quarters; each of two people gets one half.

The number of parts changes, but the total shaded amount stays the same when the wholes match.

#### Make it easier

Compare only one whole with two halves.

#### Take it further

Fold another equal sheet into eighths and find the equivalent half.

#### Another day, away from the screen

Look for halves and quarters in folded napkins.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 3. Measure twice

#### A little hint

A smaller unit usually fits more times into a fixed length.

#### Answer / what to notice

Measurements depend on the objects used; the shorter unit needs at least as many placements.

Equal units placed without gaps make a repeatable measurement; unit names make the number meaningful.

#### Make it easier

Measure a short book edge using one unit.

#### Take it further

Compare both measurements with a ruler and discuss leftover fractions.

#### Another day, away from the screen

Measure three safe objects using one shared unit and order their lengths.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 4. Timetable detective

#### A little hint

Reading ends at 3:55; five more minutes reach 4:00.

#### Answer / what to notice

Reading ends at 3:55 and drawing at 4:05; five drawing minutes come before 4:00.

The total is 25 minutes, and 20 minutes take the clock from 3:40 to 4:00.

#### Make it easier

Start at 3:00 so both activities finish within the same hour.

#### Take it further

Add a 7-minute break and find the new ending time.

#### Another day, away from the screen

Plan two real activities and compare estimated with actual durations.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 5. A tiny survey

#### A little hint

Check that all bar counts add up to the number of votes.

#### Answer / what to notice

The sample votes give drawing 3, stories 2 and building 1; drawing leads building by 2.

A chart summarizes the people or sample surveyed, not everyone everywhere.

#### Make it easier

Compare only two choices.

#### Take it further

Show the same data in a table and explain which display is easier to read.

#### Another day, away from the screen

Survey a different harmless preference and compare the totals.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 6. The value of the digit four

#### A little hint

The rightmost digit counts ones.

#### Answer / what to notice

Four tens, worth forty.

46 is  $40 + 6$ , so the 4 has value 40.

#### Make it easier

Draw four bundles of ten and six dots.

#### Take it further

What value does 4 have in 64?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 7. Cross the next ten

#### A little hint

Use three of the eight to reach thirty.

#### Answer / what to notice

35 books.

$27 + 3 = 30$ , with five left to add;  $30 + 5 = 35$ .

#### Make it easier

Use a number line and jump first to thirty.

#### Take it further

Explain a different way to calculate  $27 + 8$ .

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 8. Return the borrowed books

#### A little hint

Subtract ten first, then seven.

#### Answer / what to notice

25 books.

$42 - 10 = 32$ ;  $32 - 7 = 25$ . Check:  $25 + 17 = 42$ .

#### Make it easier

Count up from seventeen to forty-two on paper.

#### Take it further

If five more books return, how many remain borrowed?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 9. Rows in the mini theatre

#### A little hint

You can add six four times.

#### Answer / what to notice

24 seats.

$4 \times 6 = 24$ ; each row contributes six seats.

#### Make it easier

Draw four rows of six dots.

#### Take it further

If one full row is removed, how many seats remain?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 10. Three per envelope

#### A little hint

Find the biggest group of threes that does not pass seventeen.

#### Answer / what to notice

Five full envelopes, with two cards left.

$5 \times 3 = 15$  and  $17 - 15 = 2$ ; the remainder cannot fill another envelope.

#### Make it easier

Draw and circle groups of three marks.

#### Take it further

How many envelopes are needed to hold all cards, including an incomplete one?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 11. Which quarter amount is larger?

#### A little hint

The parts are equal in size; compare how many are coloured.

#### Answer / what to notice

Three quarters is larger than one quarter.

With the same whole and denominator, three equal parts cover more than one.

#### Make it easier

Draw the two rectangles and shade the stated parts.

#### Take it further

How much more is coloured in the first rectangle?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 12. Change from a twenty token

#### A little hint

Add both costs before subtracting from twenty.

#### Answer / what to notice

Eight tokens.

The total is  $7 + 5 = 12$ ;  $20 - 12 = 8$ .

#### Make it easier

Work out the total cost as a separate step.

#### Take it further

If you instead buy two pens and the notebook, what change is due?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 13. Metres and centimetres together

#### A little hint

Replace the metre with one hundred centimetres.

#### Answer / what to notice

125 centimetres.

$100 + 25 = 125$ ; the pieces use the same unit before adding.

#### Make it easier

Show one long section marked 100 and one marked 25.

#### Take it further

Convert two metres and five centimetres into centimetres.

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 14. A border around the card

#### A little hint

There are two long sides and two short sides.

#### Answer / what to notice

22 centimetres.

$8 + 3 + 8 + 3 = 22$ . Perimeter measures the boundary, not the surface inside.

#### Make it easier

Label all four sides before adding.

#### Take it further

What is the perimeter if the width becomes four centimetres?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 15. Balance the fruit baskets

#### A little hint

Both sides total five hundred grams.

#### Answer / what to notice

300 grams.

$200 + 300 = 500$ , so  $500 - 200 = 300$  grams remain for the pear.

#### Make it easier

Replace the numbers by five equal units and two known units.

#### Take it further

If the apple weighed 180 grams on the same balance, what would the pear weigh?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 16. The workshop finish time

#### A little hint

Twenty minutes takes you to three o'clock.

#### Answer / what to notice

3:15.

20 minutes reach 3:00; 15 minutes remain, giving 3:15.

#### Make it easier

Mark 2:40, 3:00 and the remaining jump on a line.

#### Take it further

What time is a ten-minute break over if it starts at 3:15?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 17. Totals from two reading days

#### A little hint

Use only the two named days.

#### Answer / what to notice

17 pages.

$8 + 9 = 17$ ; Tuesday is not part of this question.

#### Make it easier

Underline the two relevant numbers.

#### Take it further

Find the total for all three days.

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 18. Even plus even

#### A little hint

Both teams already split into complete pairs.

#### Answer / what to notice

Yes; fourteen players make seven pairs.

Adding three existing pairs and four existing pairs gives seven pairs, with no unpaired player.

#### Make it easier

Draw the pairs for each team separately.

#### Take it further

What happens if one extra player joins?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 19. Halfway between two numbers

#### A little hint

The whole gap is eight; use half of it.

#### Answer / what to notice

14.

14 is four more than 10 and four less than 18.

#### Make it easier

Draw every number from ten through eighteen.

#### Take it further

Find the midpoint between 20 and 30.

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 20. Round to the nearest ten

#### A little hint

Compare the distance from 67 to each choice.

#### Answer / what to notice

70.

67 is three away from 70 and seven away from 60, so 70 is nearer.

#### Make it easier

Mark sixty, sixty-seven and seventy on a line.

#### Take it further

Round 62 to the nearest ten and explain.

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.